Cassie Davidson
Final Lab Report

Effects of Cannabis, Methamphetamine, and their Interactions on Circadian Rhythm Activity in *Drosophila melanogaster*

Abstract:

This study aimed to examine the effects of cannabis and methamphetamine exposure on the locomotor activity and circadian rhythm of *Drosophila melanogaster*. Flies were assigned to four treatment groups: Control, Cannabis, Methamphetamine, and Cannabis + Methamphetamine. Their locomotor activity was monitored over nine days using Drosophila Activity Monitors and statistical analyses were conducted to assess the main effects of sex, cannabinol, methamphetamine, as well as their interactions on various activity parameters. It was observed that the impact of cannabinol, methamphetamine, sex, and their interactions on the survivability and circadian rhythm of fruit flies was minimal with the strongest evidence being the effect of cannabinol on mean activity of the flies in 12:12 Light:Dark cycle.

Introduction:

The study of methamphetamine and cannabinol is of considerable significance due to their profound impacts on human health. Methamphetamine, a powerful central nervous system stimulant, is known for its potential to cause addiction and severe neurological damage. It alters dopamine transmission, leading to heightened activity levels, reduced need for sleep, and disrupted circadian rhythms. Understanding its effects at a fundamental biological level can provide insights into the mechanisms of addiction, the potential for long-term neurological impacts, and possible therapeutic interventions.

Cannabinol, a non-psychoactive component of cannabis, plays a crucial role in regulating mood, sleep, and circadian rhythms. As cannabis use becomes more widespread and accepted for medicinal purposes, it is essential to understand both its therapeutic potential and any adverse effects. Studying cannabinol can help elucidate its role in modulating neurological functions and its interactions with other substances, such as methamphetamine.

In recent years, there has been growing interest in understanding the effects of various substances on biological processes, including circadian rhythm, which plays a crucial role in regulating many physiological functions. The fruit fly, *Drosophila melanogaster* (*D. melanogaster*), has long been a basis for genetic and behavioral research due to its rapid life cycle, ease of maintenance, and similarity to humans (Tolwinski). *D. melanogaster*'s protein coding region shares up to 70% similarity with humans, making them an ideal model for studying genetic processes.

This experiment investigates the effects of cannabinol and methamphetamine on the locomotor activity and circadian rhythms of *D. melanogaster*. Specifically, the study examines how these substances, individually and in combination, influence activity levels, sleep patterns, and free-running circadian periods in a controlled laboratory setting. The objective is to understand how these substances alter behavioral patterns, which can provide insights into their broader impacts on neurological functions and circadian biology. The findings are expected to contribute to a better understanding of drug interactions, potential therapeutic benefits, and risks associated with these substances.

Methods & Materials:

We utilized an outbred strain of *Drosophila melanogaster*, specifically the "Canton-S" Wild-Type strain. Flies were reared under controlled conditions at 25°C with a 12:12 Light:Dark photoperiod, where lights were turned on at 8:00 AM and off at 8:00 PM. Their diet consisted of Carolina Instant Fruit Fly Medium, composed of 5 ml food and 11 ml water, seeded with two grains of Baker's yeast. Flies were housed in polypropylene vials, with four males and four females per vial, and were mated two weeks before the experiment. Parents were removed after seven days. Offspring aged 1-4 days were anesthetized with FlyNap and placed into Drosophila Activity Monitors (Trikinetics, Inc., Waltham, MA) for three days in the 12:12 Light:Dark photoperiod, followed by six days in constant darkness in Percival insect incubators.

Each fly was housed individually in plastic tubes measuring 5 mm in diameter and 60 mm in length, containing approximately one cm of agar food (2% agar and 5% sucrose per 100 ml deionized water), sealed with a plastic cap and a cotton plug. Flies were arranged in the monitors in repeated blocks, ensuring equal exposure to environmental variation within the incubator.

The treatment groups included Control (standard agar food), Cannabinol (0.1% solution in methanol), Methamphetamine Hydrochloride (0.1% in standard agar), and a combination of Cannabinol and Methamphetamine. Data recording began at midnight, with the number of photobeam crossings per 10-minute interval recorded for each fly for the next 72 hours in the 12:12 Light:Dark photoperiod, followed by six days in constant darkness, all maintained at 25°C.

Variables analyzed for each fly included Average Activity Counts per minute in 12:12 Light:Dark (XLD), Average Activity Counts per minute in constant darkness for days four through eight (XDD), Free-running circadian period in constant darkness (TAUDD), Average sleep score per minute in 12:12 Light:Dark (SLD), Average sleep score per minute in constant darkness (SDD), Free-running circadian period averaged over six daily cycles of sleep (TAUDDS), and the number of days each fly survived over the nine-day monitoring period (DAYS).

Data analysis was conducted using the CLOCKLAB program to determine mean activity levels in one-minute intervals during the 48-hour 12:12 Light:Dark cycle. Statistical analysis was performed using the Statistical Analysis System (SAS), with an analysis of variance (ANOVA)

testing for main treatment effects (sex, cannabis, and methamphetamine) and their interactions. Results were presented as means +/- SEM, with statistical significance set at $p < 0.05$. Multiple comparison procedures were employed to identify differences among treatments when significant effects were detected in the ANOVA. The ANOVA tested the main effects of sex, cannabis, and methamphetamine, as well as their interactions. Only flies that survived for all nine days were included in the data analysis, with the "DAYS" variable encompassing all flies that lived at least one day to assess mortality effects.

Results:

Tables 1-4 provide the means and standard error of the mean for different combinations of variables and treatment groups. Table 1 provides the means and for all variables for all flies. Table 2 provides the averages and SEM between the sexes. Table 3 shows the differences in the averages and SEM between the different treatment groups and Table 4 provides the differences between the averages and SEM of each treatment group and between the sexes for all variables.

Looking at Table 2, there are some statistical differences between the sexes, specifically in the XLD. The females have an average of 0.925 +/- 0.069 counts per minute, while the males have an average of 1.190 +/- 0.104 counts per minute. These numbers with their SEMs do not overlap suggesting a possible statistical significant difference between the two. This is a very small difference and there were 45 male flies tested compared to the 37 females, so it is possible this is due to the sample size or there could be a difference in female versus males in their average activity counts for this variable.

Table 3 shows flies that were given methamphetamine (MA) or a combination of methamphetamine and cannabinol (MA+CA) did not have a 100% survival rate, while flies not given methamphetamine did have a 100% survival rate. The top left in table 3 shows the control flies having a mean survival of exactly 9 days +/- 0. The bottom left in table 3 shows the CA flies having the same mean and SEM of 9 +/- 0. The top right table shows the MA group having a mean of 8.550 +/- 0.328 and the bottom right shows the MA+CA flies having an average survival of 8.240 +/- 0.437 days. Due to no overlap between MA and MA+CA and the control and CA, this suggests there is likely a statistical significance between these treatments and the control and CA. This data suggests that methamphetamine and the combination between methamphetamine and cannabinol may have an effect on survivability of the flies.

Table 3 also provides insight on the average activity counts per minute in a 12:12 Light:Dark cycle (XLD). The control flies had an average of 1.576 +/- 0.191. The CA flies had an average of 0.827 +/- 0.070 and the MA+CA flies had an average of 0.801 +/- 0.072. Due to no overlap between CA and MA+CA and the control, this suggests there is likely a statistical significance between these treatments and the control. This data suggests that cannabinol and the combination between cannabinol and methamphetamine may have an effect on the sleep cycle. Similarly, looking at the SLD (average sleep score per minute in 12:12 Light:Dark cycle), there are some possible significant findings with the CA flies. Their mean SLD is 59.7% +/- 02.5%

asleep and the controls mean was 35.3% +/- 3.02% time asleep. Additionally, the MA+CA had a mean of 56.4% +/-2.9%. Both the means of the CA and MA+CA do not overlap with the control's average suggesting that cannabinol and its interaction with methamphetamine may have an effect on the percentage of sleep of the flies.

Variable	N	Mean	Std Error
XLD	82	1.0704390	0.0662532
XDD	82	1.3181463	0.0543503
TAUDD	67	24.6291045	0.0638465
SLD	82	0.5163415	0.0181512
SDD	82	0.3063293	0.0159293
TAUDDS	73	24.6369863	0.0564017
DAYS	87	8.6781609	0.1484971

Table 1. The mean +/- Standard Error of the mean (SEM) for each variable for all flies.

SEX=F

Variable	N	Mean	Std Error
XLD	37	0.9248378	0.0687807
XDD	37	1.2247838	0.0820402
TAUDD	32	24.5562500	0.0848608
SLD	37	0.4821622	0.0231528
SDD	37	0.2779459	0.0211766
TAUDDS	36	24.5638889	0.0828943
DAYS	41	8.4634146	0.2773955

SEX=M

Variable	N	Mean	Std Error
XLD	45	1.1901556	0.1040208
XDD	45	1.3949111	0.0713056
TAUDD	35	24.6957143	0.0941876
SLD	45	0.5444444	0.0265483
SDD	45	0.3296667	0.0228490
TAUDDS	37	24.7081081	0.0759453
DAYS	46	8.8695652	0.1304348

Table 2. The means +/- SEM for each variable for each sex for all flies.

MA=N CA=N

Variable	N	Mean	Std Error
XLD	14	1.5758571	0.1907015
XDD	14	1.5598571	0.1788022
TAUDD	12	24.6166667	0.0898933
SLD	14	0.3526429	0.0301501
SDD	14	0.1947143	0.0230099
TAUDDS	13	24.5769231	0.0976676
DAYS	14	9.0000000	0

MA=Y CA=N

Variable	N	Mean	Std Error
XLD	18	1.3841667	0.1604976
XDD	18	1.5342778	0.0792773
TAUDD	13	24.6038462	0.1946777
SLD	18	0.4590556	0.0400155
SDD	18	0.2528889	0.0307979
TAUDDS	17	24.6970588	0.1225848
DAYS	20	8.5500000	0.3282730

MA=N CA=Y

Variable	N	Mean	Std Error
XLD	28	0.8273929	0.0697886
XDD	28	1.0880357	0.0786974
TAUDD	26	24.7115385	0.0762237
SLD	28	0.5972500	0.0253209
SDD	28	0.3783214	0.0260455
TAUDDS	27	24.7018519	0.0652347
DAYS	28	9.0000000	0

MA=Y CA=Y

Variable	N	Mean	Std Error
XLD	22	0.8014545	0.0717446
XDD	22	1.2803636	0.0933211
TAUDD	16	24.5250000	0.1696811
SLD	22	0.5644091	0.0286729
SDD	22	0.3294545	0.0295424
TAUDDS	16	24.5125000	0.1789728
DAYS	25	8.2400000	0.4369592

Table 3. The means +/- SEM for each variable for each treatment group for all flies.

SEX=F MA=N CA=N

Variable	N	Mean	Std Error
XLD	7	1.1134286	0.2042572
XDD	7	1.2591429	0.2487748
TAUDD	7	24.5428571	0.1043281
SLD	7	0.3581429	0.0498667
SDD	7	0.2382857	0.0263906
TAUDDS	7	24.6071429	0.1320302
DAYS	7	9.0000000	0

SEX=F MA=Y CA=N

Variable	N	Mean	Std Error
XLD	6	1.2348333	0.1593603
XDD	6	1.5675000	0.0667901
TAUDD	3	24.5833333	0.2803767
SLD	6	0.4040000	0.0638733
SDD	6	0.1585000	0.0257303
TAUDDS	5	24.6900000	0.2972373
DAYS	7	8.5714286	0.4285714

SEX=M MA=N CA=N

Variable	N	Mean	Std Error
XLD	7	2.0382857	0.2111625
XDD	7	1.8605714	0.2156517
TAUDD	5	24.7200000	0.1609348
SLD	7	0.3471429	0.0379783
SDD	7	0.1511429	0.0310648
TAUDDS	6	24.5416667	0.1572772
DAYS	7	9.0000000	0

SEX=M MA=Y CA=N

Variable	N	Mean	Std Error
XLD	12	1.4588333	0.2289605
XDD	12	1.5176667	0.1161052
TAUDD	10	24.6100000	0.2458319
SLD	12	0.4865833	0.0507747
SDD	12	0.3000833	0.0379592
TAUDDS	12	24.7000000	0.1325736
DAYS	13	8.5384615	0.4615385

SEX=F MA=N CA=Y

Variable	N	Mean	Std Error
XLD	14	0.7645000	0.0770603
XDD	14	0.9800714	0.0998631
TAUDD	13	24.7807692	0.1177430
SLD	14	0.5290714	0.0254140
SDD	14	0.3350000	0.0369382
TAUDDS	14	24.7571429	0.0785964
DAYS	14	9.0000000	0

SEX=F MA=Y CA=Y

Variable	N	Mean	Std Error
XLD	10	0.8313000	0.1262065
XDD	10	1.3377000	0.1744232
TAUDD	9	24.2333333	0.1846543
SLD	10	0.5502000	0.0405816
SDD	10	0.2975000	0.0413498
TAUDDS	10	24.2000000	0.1854424
DAYS	13	7.5384615	0.8055527

SEX=M MA=N CA=Y

Variable	N	Mean	Std Error
XLD	14	0.8902857	0.1169798
XDD	14	1.1960000	0.1181524
TAUDD	13	24.6423077	0.0977054
SLD	14	0.6654286	0.0360865
SDD	14	0.4216429	0.0341274
TAUDDS	13	24.6423077	0.1066456
DAYS	14	9.0000000	0

SEX=M MA=Y CA=Y

Variable	N	Mean	Std Error
XLD	12	0.7765833	0.0838554
XDD	12	1.2325833	0.0963382
TAUDD	7	24.9000000	0.2537809
SLD	12	0.5762500	0.0415521
SDD	12	0.3560833	0.0417976
TAUDDS	6	25.0333333	0.2574447
DAYS	12	9.0000000	0

Table 4. The mean +/- SEM for each variable and each sex within each treatment group for all flies.

Based on the results from the GLM analysis, TAUDD, TAUDDS, and DAYS are not a good fit for the model. The general null hypotheses being that there is no statistical difference between the factors (cannabinol, methamphetamine, sex, and their interactions) for each of the dependent variables (XLD, XDD, TAUDD, SLD, SDD, TAUDDS, DAYS). The alternative hypothesis is that there is a statistical difference between at least one of the factors for the dependent variable. TAUDD and DAYS show no statistical significance between any of the treatment groups, supporting the null hypotheses. In Table 7, the F-value of 1.25 and the p-value of 0.2909 for TAUDD indicate that none of the variables are significant. Additionally, the low R-squared value of 0.1291 suggests that the model explains only about 13% of the total variability in the data. Similarly, in Table 11, the F-value of 1.81 and the p-value of 0.0967 for DAYS also indicate non-significance. The R-squared value of 0.1383 further supports this, indicating that the model explains only about 14% of the total variability. The larger p-values for both TAUDD and DAYS analyses suggest no statistical difference between the variables.

In contrast, Table 10 (left) reveals an F-value of 2.18 and a p-value of 0.0476 for TAUDDS. While the p-value is less than the standard 0.05, indicating statistical significance, the R-squared value of 0.190 suggests that the model explains only about 19% of the total variability. For all three variables (TAUDD, DAYS, TAUDDS), the low R-squared values indicate a poor fit of the model to the data, suggesting the presence of other unaccounted factors contributing to the variability.

XLD, SLD, and SDD are all good fits, with their data better representing the model. Although XDD is a better fit than the first three variables, it still does not provide an ideal fit for the data. Table 6 (left) shows XDD with an R-squared value of 0.246, indicating that the model explains about 24.6% of the variance. The statistically significant p-value (0.0029) suggests that the factors in the model do have a significant effect on XDD, despite the modest overall fit. The data in Table 6 (right) shows statistically significant values for CA (F-value = 12.74, p-value = 0.0006) and SEX*MA (F-value = 5.18, p-value = 0.0287), suggesting limited evidence of the

effect of cannabinol and the interaction between sex and methamphetamine on mean activity counts per minute in constant darkness for days four through eight.

The SLD and SDD yield similar results, both showing strong statistical significance. Table 8 (left) indicates a very strong statistical significance for SLD, with an F-value and p-value of 6.46 and <0.0001 respectively, and an R-squared value of 0.379. Similarly, Table 9 (left) shows strong statistical significance for SDD, with an F-value and p-value of 5.47 and <0.0001 respectively, and an R-squared value of 0.332. Table 8 (right) and Table 9 (right) indicate significant differences in CA and SEX interactions, suggesting effects on sleep scores.
Table 5 (left) contains an F-value of 7.31 and a p-value of <0.0001 for XLD, indicating a statistically significant difference in the factors. This F-value, along with an R-squared value of 0.409, is larger than all the other variables, further demonstrating the goodness of fit for XLD. The p-value being less than the standard 0.05 shows there is strong evidence to reject the null hypothesis that there is no relationship between these three factors and the flies' average activity counts per minute.

Table 5 (left) contains an F-value of 7.31 and a p-value of <0.0001 for XLD, indicating a statistically significant difference in the factors. This F-value, along with an R-squared value of 0.409, is larger than all the other variables, further demonstrating the goodness of fit for XLD. The p-value being less than the standard 0.05 shows there is strong evidence to reject the null hypothesis that there is no relationship between these three factors and the flies' average activity counts per minute. Table 5 (right) shows that CA has a large effect on XLD, with a p-value of <0.0001 and an F-value of 33.68. CA and SEX, along with their interaction (SEX*CA), have statistically significant differences compared to other factors.

Dependent Variable: XLD

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	11.91522518	1.70217503	7.31	<.0001
Error	74	17.23976102	0.23296974		
Corrected Total	81	29.15498620			

R-Square	Coeff Var	Root MSE	XLD Mean
0.408686	45.09079	0.482669	1.070439

Source	DF	Type I SS	Mean Square	F Value	Pr > F
SEX	1	1.42933126	1.42933126	6.14	0.0155
CA	1	7.84659438	7.84659438	33.68	<.0001
SEX*CA	1	0.99767985	0.99767985	4.28	0.0420
MA	1	0.31700163	0.31700163	1.36	0.2472
SEX*MA	1	0.74899549	0.74899549	3.21	0.0771
CA*MA	1	0.26147060	0.26147060	1.12	0.2929
SEX*CA*MA	1	0.31415197	0.31415197	1.35	0.2493

Table 5. Analysis of mean activity in 12:12 Light:Dark cycle (XLD) and results from the GLM procedure.

Dependent Variable: XDD

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	4.83538556	0.69076937	3.46	0.0029
Error	74	14.78476468	0.19979412		
Corrected Total	81	19.62015024			

R-Square	Coeff Var	Root MSE	XDD Mean
0.246450	33.90999	0.446983	1.318146

Source	DF	Type I SS	Mean Square	F Value	Pr > F
SEX	1	0.58769033	0.58769033	2.94	0.0905
CA	1	2.54576666	2.54576666	12.74	0.0006
SEX*CA	1	0.12016954	0.12016954	0.60	0.4405
MA	1	0.16293543	0.16293543	0.82	0.3694
SEX*MA	1	1.03574653	1.03574653	5.18	0.0257
CA*MA	1	0.25656304	0.25656304	1.28	0.2608
SEX*CA*MA	1	0.12651403	0.12651403	0.63	0.4287

Table 6. Analysis of mean activity counts per minute in constant darkness for days four through eight in the monitors (XDD) and results from the GLM procedure.

Dependent Variable: TAUDD

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	2.32801367	0.33257338	1.25	0.2909
Error	59	15.69773260	0.26606326		
Corrected Total	66	18.02574627			

R-Square	Coeff Var	Root MSE	TAUDD Mean
0.129149	2.094324	0.515813	24.62910

Table 7. Analysis of the "Free Running Circadian Period" in constant darkness (TAUDD).

This represents the average duration of one complete circadian cycle, measured in hours, observed over six consecutive days in constant darkness.

Dependent Variable: SLD

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	0.82991360	0.11855909	6.46	<.0001
Error	74	1.35840084	0.01835677		
Corrected Total	81	2.18831444			

R-Square	Coeff Var	Root MSE	SLD Mean
0.379248	26.23984	0.135487	0.516341

Source	DF	Type I SS	Mean Square	F Value	Pr > F
SEX	1	0.07876430	0.07876430	4.29	0.0418
CA	1	0.59996851	0.59996851	32.68	<.0001
SEX*CA	1	0.00443396	0.00443396	0.24	0.6245
MA	1	0.00486916	0.00486916	0.27	0.6081
SEX*MA	1	0.00340622	0.00340622	0.19	0.6679
CA*MA	1	0.09023952	0.09023952	4.92	0.0297
SEX*CA*MA	1	0.04823193	0.04823193	2.63	0.1093

Table 8. Analysis of the average sleep score per minute in 12:12 (SLD) and results from the GLM procedure.

Dependent Variable: SDD

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	0.56073278	0.08010468	5.27	<.0001
Error	74	1.12463133	0.01519772		
Corrected Total	81	1.68536411			

R-Square	Coeff Var	Root MSE	SDD Mean
0.332707	40.24396	0.123279	0.306329

Source	DF	Type I SS	Mean Square	F Value	Pr > F
SEX	1	0.05431622	0.05431622	3.57	0.0626
CA	1	0.34799281	0.34799281	22.90	<.0001
SEX*CA	1	0.00380891	0.00380891	0.25	0.6181
MA	1	0.00292780	0.00292780	0.19	0.6620
SEX*MA	1	0.02719233	0.02719233	1.79	0.1851
CA*MA	1	0.04799100	0.04799100	3.16	0.0797
SEX*CA*MA	1	0.07650370	0.07650370	5.03	0.0278

Table 9. Analysis of average sleep score per minute in constant darkness (SDD) and results from the GLM procedure.

Dependent Variable: TAUDDS

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	3.17706098	0.45386585	2.18	0.0476
Error	65	13.54307601	0.20835502		
Corrected Total	72	16.72013699			

R-Square	Coeff Var	Root MSE	TAUDDS Mean
0.190014	1.852740	0.456459	24.63699

Source	DF	Type I SS	Mean Square	F Value	Pr > F
SEX	1	0.37951386	0.37951386	1.82	0.1818
CA	1	0.00153352	0.00153352	0.01	0.9319
SEX*CA	1	0.23729867	0.23729867	1.14	0.2898
MA	1	0.03515192	0.03515192	0.17	0.6826
SEX*MA	1	1.61742077	1.61742077	7.76	0.0070
CA*MA	1	0.15050140	0.15050140	0.72	0.3985
SEX*CA*MA	1	0.75564085	0.75564085	3.63	0.0613

Table 10. Analysis of "Free Running Circadian Period" over six daily cycles of sleep (TAUDDS) and results from the GLM procedure.

Dependent Variable: DAYS

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	22.8126816	3.2589545	1.81	0.0967
Error	79	142.1758242	1.7996940		
Corrected Total	86	164.9885057			

R-Square	Coeff Var	Root MSE	DAYS Mean
0.138268	15.45865	1.341527	8.678161

Table 11. Analysis of the number of days each fly survived over nine days in the monitor (DAYS).

Discussion:

Overall, the results of the GLM procedure suggest that while there were some statistically significant findings in the models, the impact of cannabinol, methamphetamine, sex, and their interactions on the circadian rhythm activity and survivability of fruit flies was generally minimal, with the strongest data being the effect of cannabinol on mean activity in the 12:12 Light:Dark cycle.

The hypotheses tested across the various models included whether cannabinol, methamphetamine, sex, and their interactions significantly affect the activity and sleep scores of fruit flies. For TAUDD and DAYS, the lack of statistical significance and low R-squared values suggest that these factors do not have a meaningful impact on total activity under these conditions. However, for TAUDDS, the significant p-value suggests some effect, but the low R-squared value indicates that the model does not capture all relevant variability. The averages and the SEMs suggest that methamphetamine and a combination of methamphetamine and cannabinol may have an effect on the survivability of the flies.

For future investigations, it would be advantageous to explore further the ramifications of cannabinol and methamphetamine exposure on the circadian rhythm activity and survival of *D. melanogaster*. This could involve delving into additional variables or refining experimental protocols. Expanding the scope to examine the effects of these substances on other behavioral and physiological parameters could provide a more comprehensive understanding of their impact. Considering the intricate interplay between sex, cannabinol, and methamphetamine observed in this study, future research could aim to elucidate the underlying mechanisms driving these interactions and their implications for broader biological processes.

Employing refined methodologies, such as utilizing inbred fly strains, may offer benefits in terms of reducing genetic variability and enhancing result reliability. By minimizing genetic variation within experimental groups, researchers can potentially gain clearer insights into the specific effects of cannabinol and methamphetamine. However, it's essential to acknowledge that

utilizing inbred strains might limit the generalizability of findings to diverse populations or species. Therefore, a balanced approach that incorporates both inbred and outbred strains could provide a more nuanced understanding of the effects of these substances across different genetic backgrounds. This diversified approach would strengthen the robustness of conclusions drawn from future investigations into the impact of cannabinol and methamphetamine on Drosophila melanogaster circadian rhythms and behavioral patterns.

References:

Possidente, B. Biostatistics Syllabus. May 2023.

Tolwinski NS. Drosophila: A Model System for Developmental Biology. J Dev Biol.